

Land and Water Resources Engineering

- ✓ Hydrology
- ✓ Irrigation and Drainage Engineering
- ✓ Soil and Water Conservation Engineering



Engr. Stella A. Lamban

Faculty

Department of Agricultural Engineering

College of Engineering

Mindanao State University – Marawi



PROFESSIONAL REGULATION COMMISSION
PROFESSIONAL REGULATORY BOARD OF AGRICULTURAL AND BIOSYSTEMS ENGINEERING
TABLES OF SPECIFICATION



SUBJECT B. Land and Water Resources Engineering and Allied Subjects (32%)

I. Hydrology The registered and licensed Agricultural and Biosystems Engineer can perform the competencies under the following sub-topics:	Percentage Weight and Distribution	Number of Items and Distribution	Easy (30%)		Moderate (50%)	Difficult (20%)		
	32%	100	Remembering Recall information	Understanding Explain, Interpret, summarize, examine	Applying Using the information in another familiar situation	Analyzing Understanding, Comparing, organizing, deconstructing, interrogating, finding	Evaluating hypothesizing, critiquing, experimenting, judging	Creating Generating Designing, constructing, planning, producing, inventing
	2	8	Recognizing, listing, describing, retrieving, naming, finding	paraphrase, classify	Implementing carrying out, using, executing			
1. Describe the hydrologic cycle; hydrology in engineering.	0.5	2	2					
2. Describe the principles of hydrometeorology; weather and hydrology; precipitation, evaporation and evapotranspiration; weather instruments.	0.5	2	1	1				
3. Describe the principles of watershed/surface hydrology; runoff, infiltration, seepage and percolation, stream flow, soil erosion and sediment transport.	0.5	2	1	1				
4. Describe the basic theories of aquifer systems and lithography; water quality analysis.	0.5	2	2					

II. Irrigation and Drainage Engineering The registered and licensed Agricultural and Biosystems Engineer can perform the competencies under the following sub-topics:	Percentage Weight and Distribution	Number of Items and Distribution	Easy (30%)		Moderate (50%)	Difficult (20%)		
	32%	100	Remembering Recall information Recognizing, listing, describing, retrieving, naming, finding	Understanding Explain, Interpret, summarize, examine paraphrase, classify	Applying Using the information in another familiar situation Implementing carrying out, using, executing	Analyzing Understanding, Comparing, organizing, deconstructing, interrogating, finding	Evaluating hypothesizing, critiquing, experimenting, judging	Creating Generating Designing, constructing, planning, producing, inventing
	8	25						
1. Apply theories and practices in leveling, mapping, triangulation, topographic, route, and construction surveys, and global positioning system (GPS).	0.64	2	1		1			
2. Apply the theories and principles of fluid mechanics in the design, construction and management of agricultural and biosystems engineering production systems and facilities.	1.28	4	1	1	2			
3. Apply the basic theories of groundwater hydrology in planning water resource projects.	0.64	2	1		1			
4. Analyze the design, construction, operation and maintenance of hydraulics machinery and facilities (i.e. pumps, and turbines); pump size determination and specifications.	1.28	4				2	2	
5. Plan, design, construct, operate and manage irrigation and drainage systems and facilities; design small canal structures for the conveyance of irrigation or drainage water.	2.88	9		2	1	2	2	2
6. Apply the theories and principles in designing pressurized irrigation systems and facilities.	1.28	4			4			

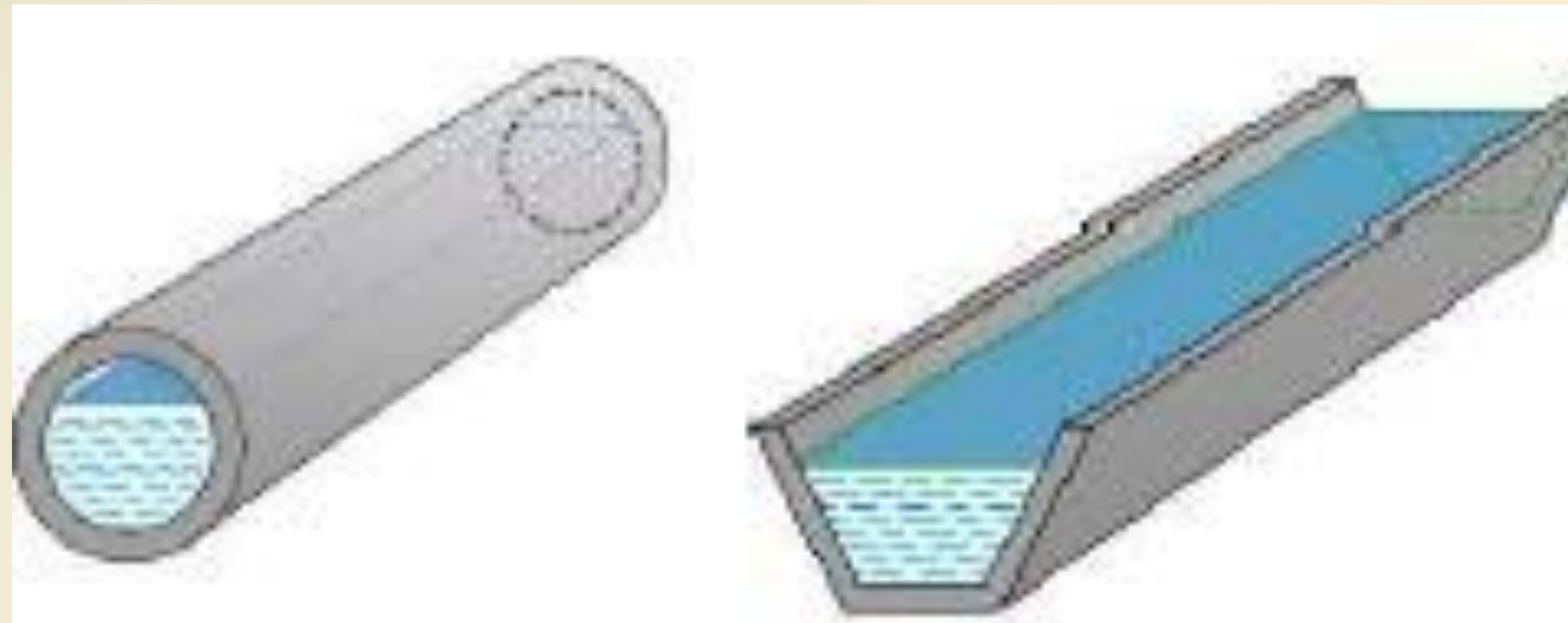
III. Soil and Water Conservation Engineering The registered and licensed Agricultural and Biosystems Engineer can perform the competencies under the following sub-topics:	Percentage Weight and Distribution	Number of Items and Distribution	Easy (30%)		Moderate (50%)	Difficult (20%)		
	32%	100	Remembering Recall information Recognizing, listing, describing, retrieving, naming, finding	Understanding Explain, Interpret, summarize, examine paraphrase, classify	Applying Using the information in another familiar situation Implementing carrying out, using, executing	Analyzing Understanding, Comparing, organizing, deconstructing, interrogating, finding	Evaluating hypothesizing, critiquing, experimenting, judging	Creating Generating Designing, constructing, planning, producing, inventing
	6	20						
1. Explain and apply the basic soil-water- plant relations in crop production.	0.6	2		1	1			
2. Apply the principles of hydrometeorology in the planning and design and management of water resources facilities.	0.30	1			1			
3. Apply the principles of watershed hydrology in the planning and design of agricultural and biosystems engineering facilities.	0.30	1			1			
4. Design, construct, manage farm ponds, reservoirs, including earth dams, and spillways.	2.4	8			4	2	1	1
5. Design of soil and water control structures and facilities for stream banks and sloping cultivated agricultural areas.	1.2	4			2		1	1
6. Design of vegetative and lined water conveyance channels.	0.9	3			1		1	1
7. Maintain water conservation works, drainage and erosion control systems.	0.3	1			1			

FLUID MECHANICS

OPEN CHANNEL HYDRAULICS (CANAL DESIGN)

Open Channel

- It has free surface subject to atmospheric pressure.
- Flow is caused by gravity component along the slope of the channel.
- The flow may either be uniform or non-uniform



Sample Question 26:

A conduit in which a liquid flows with a free surface.

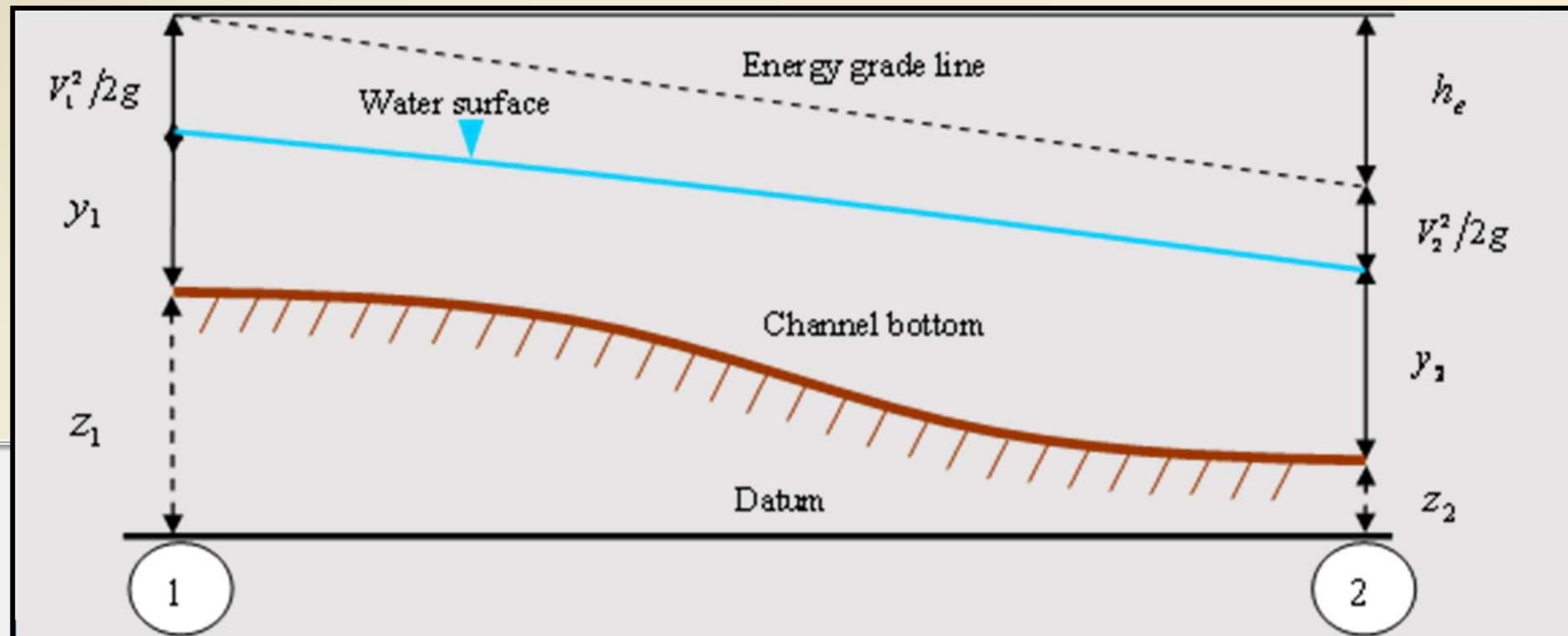
- a. Open Channel
- b. Pipe Flow
- c. Canal
- d. Waterways

A

SPECIFIC ENERGY

The specific energy (H) is defined as the energy per unit weight relative to the bottom of the channel.

$$H = \frac{v^2}{2g} + d$$



Sample Question 27:

Slope of the free water surface in open channel

- a) Energy grade line slope
- b) Water surface slope
- c) Channel bottom slope
- d) Hydraulic grade line slope

Sample Question 28:

Slope of the water surface profile plus the velocity head in open channels

- a) Energy grade line slope
- b) Water surface slope
- c) Channel bottom slope
- d) Hydraulic grade line slope

Chezy's Equation

$$V = C\sqrt{RS}$$

$$Q = AC\sqrt{RS}$$

where:

C = Chezy coefficient

*(30 m^{1/2}/s for small rough channels to
90 m^{1/2}/s for large smooth channels)*

Kutter - Ganguillet

$$c = \frac{23 + \frac{0.00155}{S} + \frac{1}{n}}{1 + \frac{n}{\sqrt{R}} \left\{ 23 + \frac{0.00155}{S} \right\}}$$

Manning

$$c = \frac{R^{1/6}}{n}$$

Bazin

$$c = \frac{87}{1 + \frac{m}{\sqrt{R}}}$$

Manning's Equation

$$V = \frac{k}{n} R^{2/3} S^{1/2}$$

***V** = velocity of flow (m/s or ft/s)*

***R** = hydraulic radius (m or ft)*

***k** = conversion factor (**1**-metric, **1.486**-english)*

***n** = roughness coefficient*

***S** = slope of energy grade line*



UNIFORM FLOW

For the flow to be uniform, the velocity, depth of flow, and cross-sectional area of flow at any point of the stream must be constant

Normal depth d_n – the depth at which uniform flow will occur in an open channel.



BOUNDARY SHEAR STRESS

The average boundary shear stress, τ_0 , acting over the wetted surface of the channel is given by:

$$\tau_0 = \gamma RS$$

γ = unit weight of the liquid

R = hydraulic radius

S = slope of energy grade line



Sample Question 29:

Occurs when flow has a constant water area, depth, discharge, and average velocity through a reach of channel

- a. Normal flow
- b. Critical flow
- c. Uniform flow
- d. Varied flow

c

Sample Question 30:

Constant flow depth along a longitudinal section of a channel under a uniform flow condition

- a. Normal depth
- b. Critical depth
- c. Uniform depth
- d. None of the above

A

Canal Design

Non-erodible (Lined Canal)



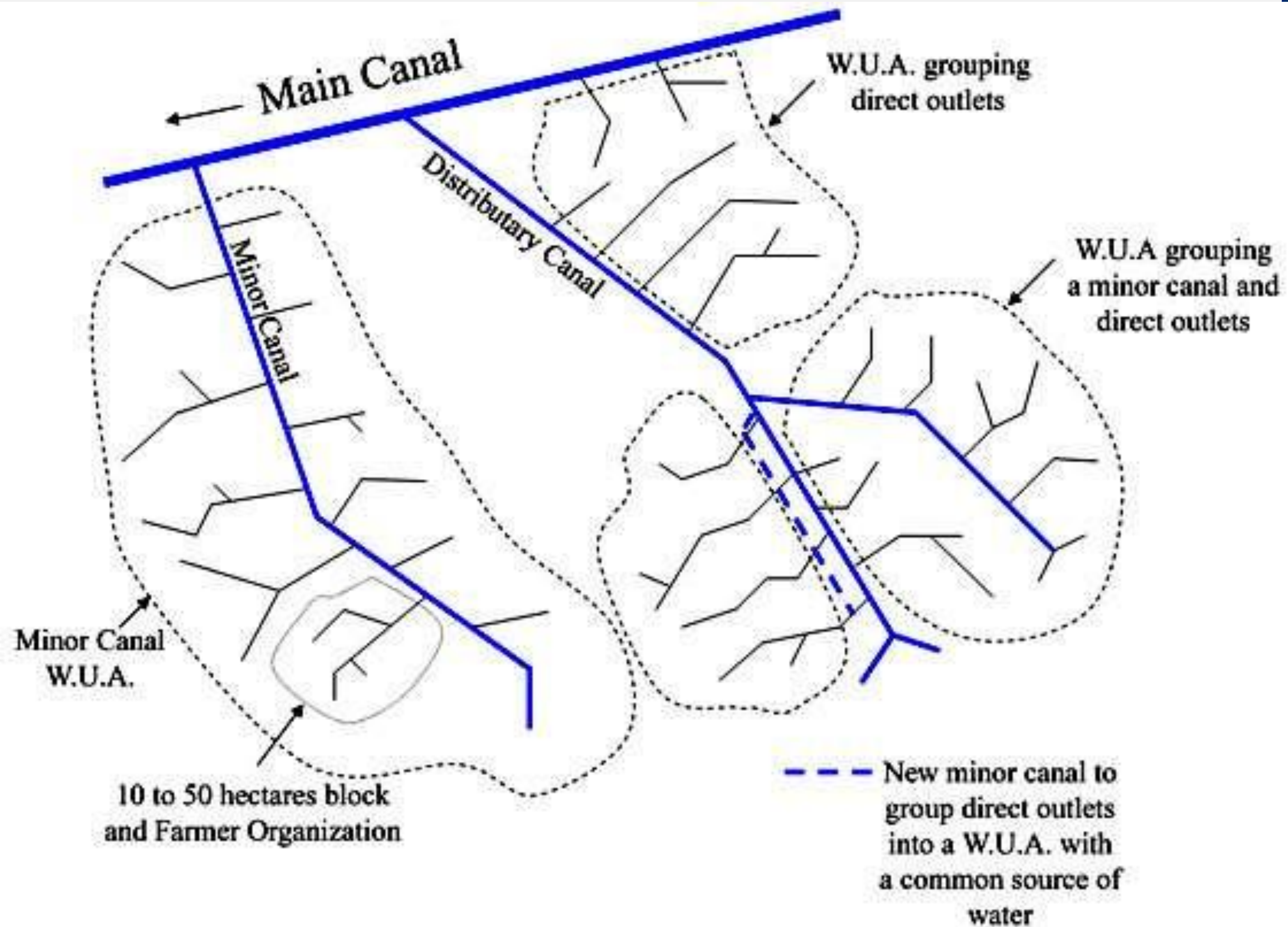
Erodible (Unlined Canal)



Design Considerations

1. Design Discharge, Q
2. Roughness coefficient, n
3. Side Slope, z
4. Bottom Slope, S
5. Minimum Permissible Velocity
6. Critical Depth
7. Free Board





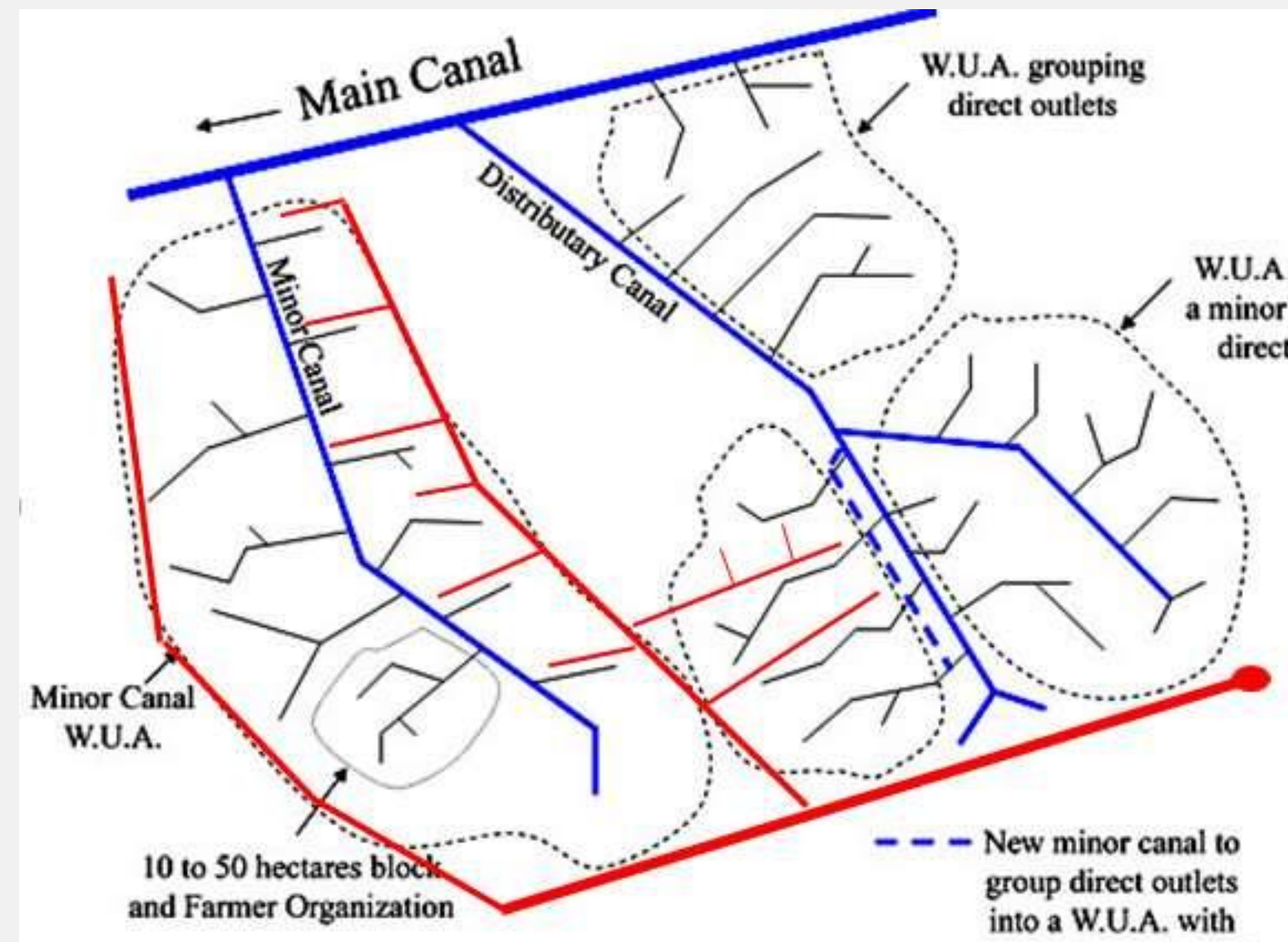
1. Design Discharge (Q)

IRRIGATION

Main Canal –
Diversion Water Requirement

DRAINAGE

Drainage Requirement



2. roughness coefficient (n)

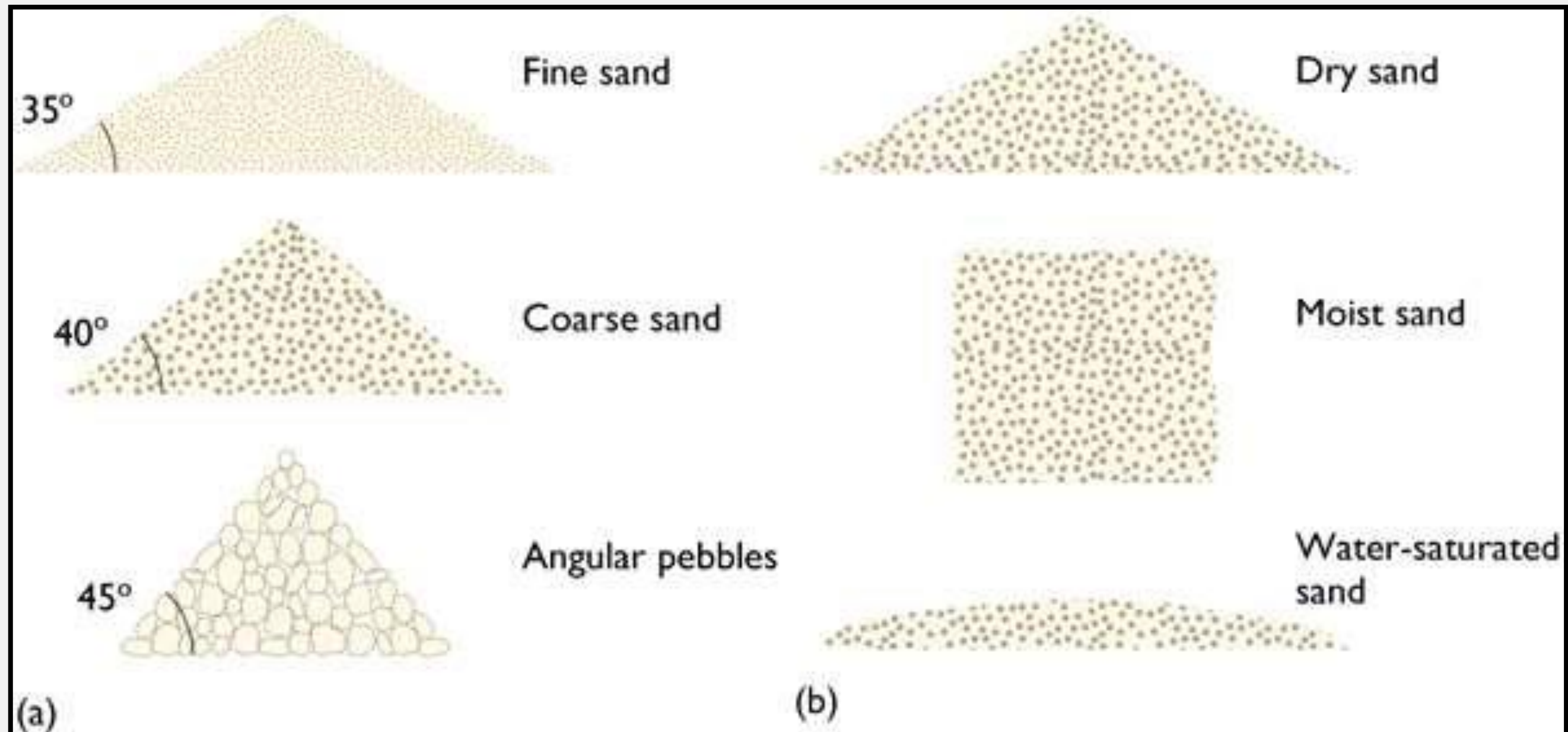
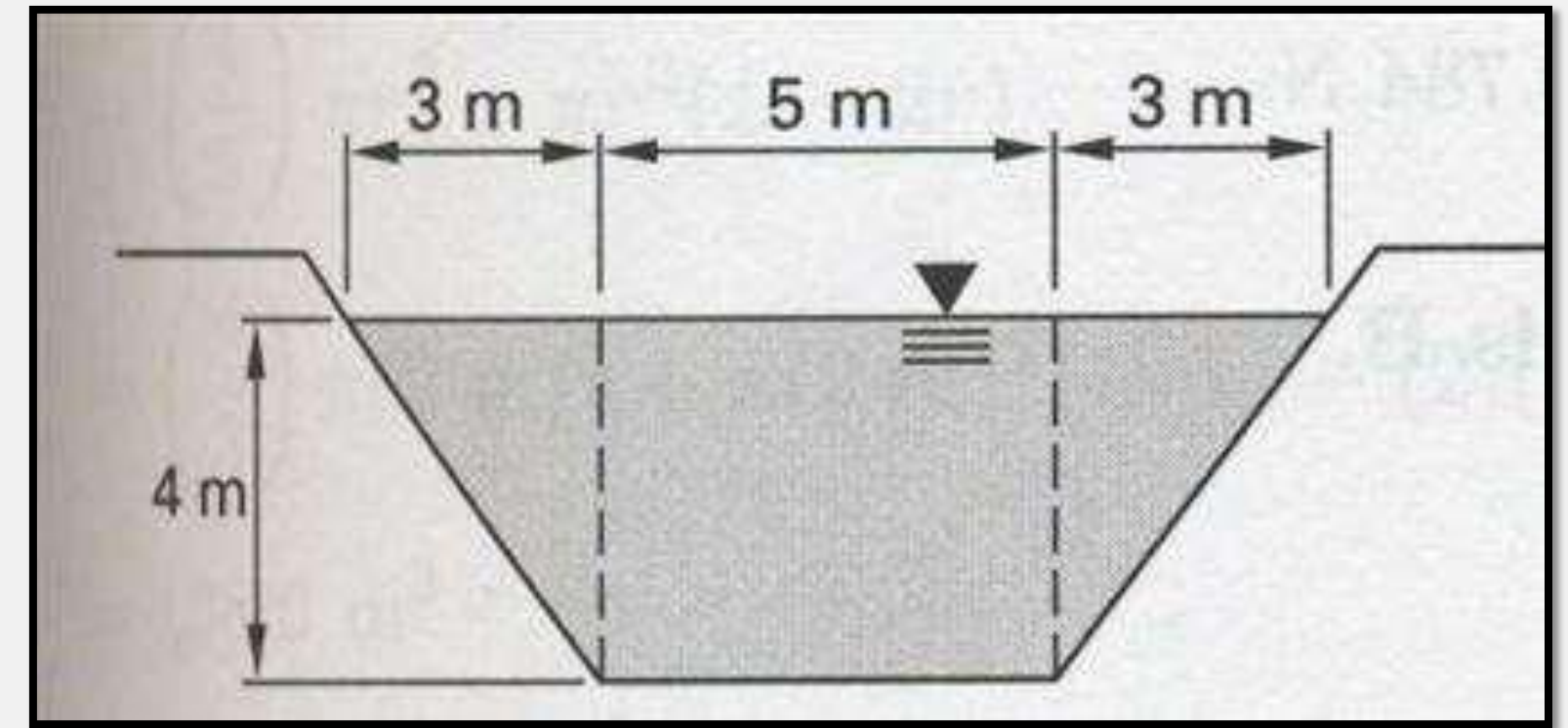
- nature of materials forming the channel



<u>Channel Surface</u>	<u>Manning Roughness Coefficient, n</u>
Asbestos cement	0.011
Brass	0.011
Brick	0.015
Cast-iron, new	0.012
Concrete, steel forms	0.011
Concrete, wooden forms	0.015
Concrete, centrifugally spun	0.013
Copper	0.011
Corrugated metal	0.022
Galvanized Iron	0.016
Lead	0.011
Plastic	0.009
Steel - Coal-tar enamel	0.01
Steel - New unlined	0.011
Steel - Riveted	0.019
Wood stave	0.012

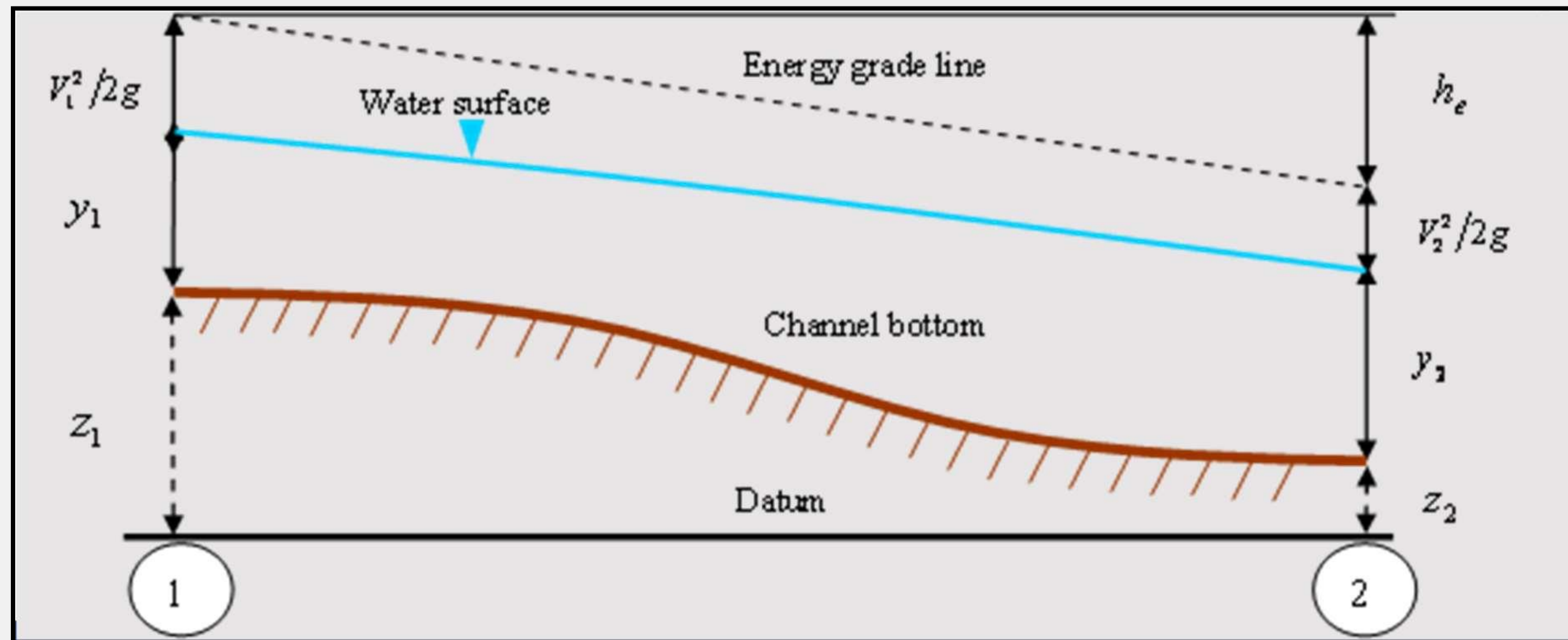
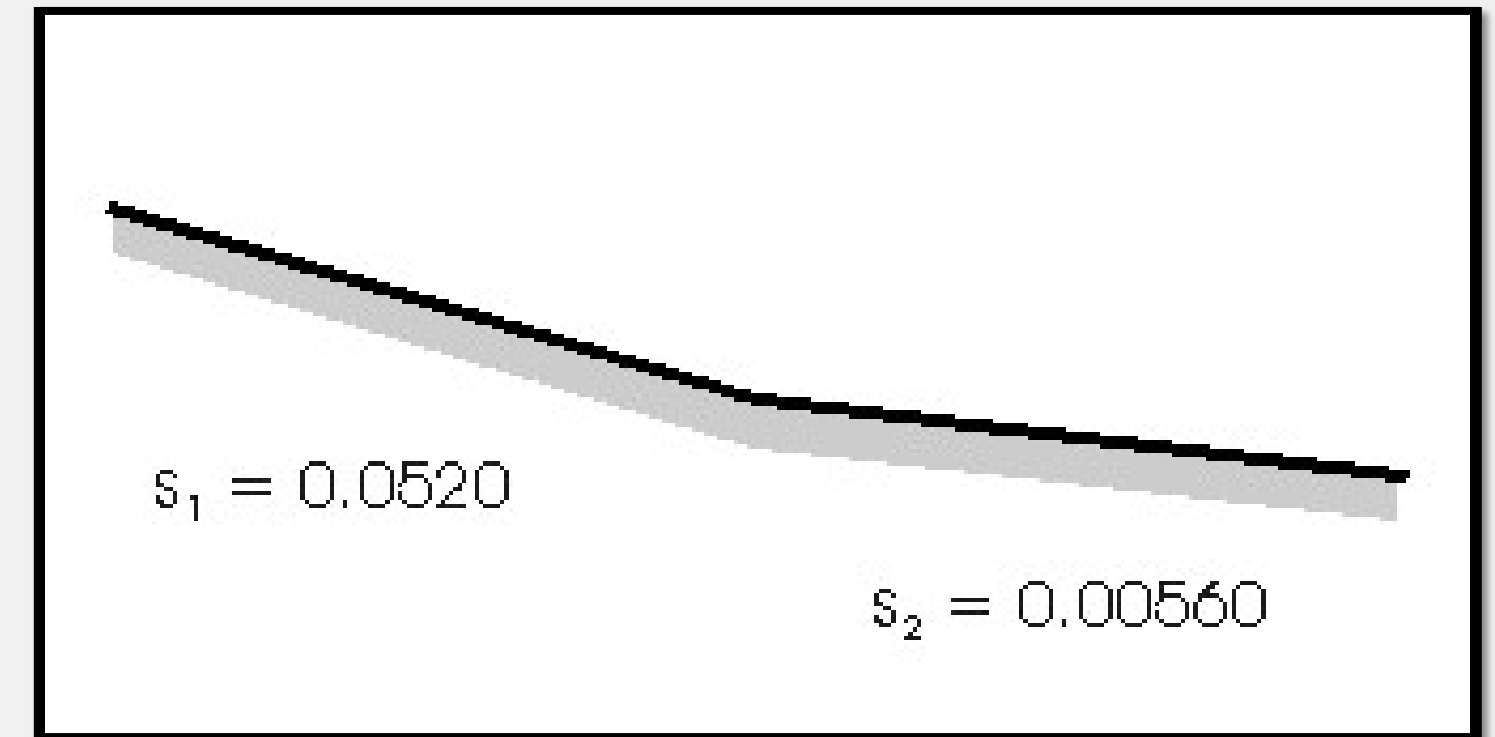
3. side slope(z)

- based on the angle of repose property of the soil



4. bottom slope (S)

- topography and head loss consideration



5. minimum permissible velocity

- for non-erodible channels, the velocity through the canal must be at least

0.6 to 0.9 m/s
(0.75 m/s)



6. critical depth (d_c)

➤ to avoid hydraulic jump along the channel

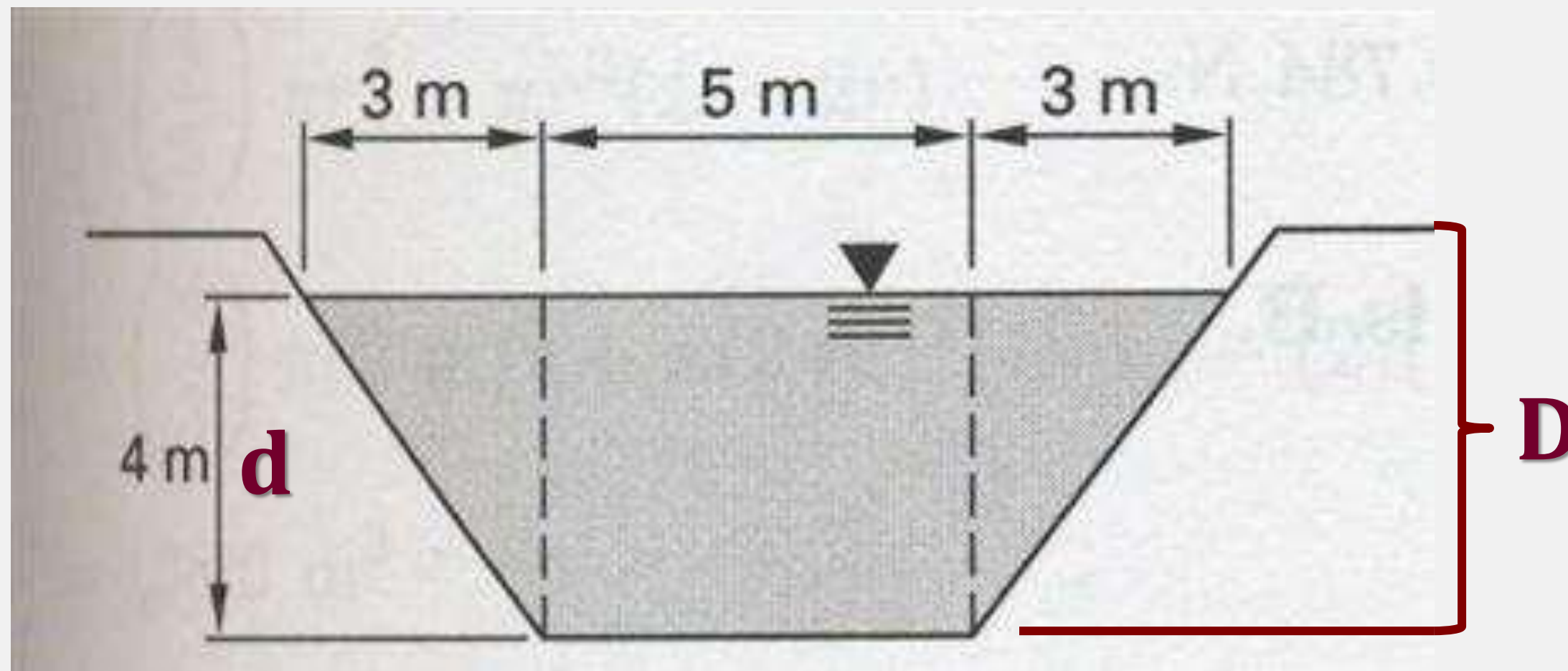
$$Fr = \frac{V}{\sqrt{gD}}$$

$$d \neq d_c$$



7. Free Board (FB)

- depth allowance to avoid spillage caused by wave action



$$FB = D - d$$



8. Other Considerations

$$Fr \leq 0.80$$

$V_{\max} = 2.1 \text{ m/s}$ (for non-reinforced canal linings)

$V_{\max} = 5.5 \text{ m/s}$ (for reinforced canal linings)

$V_{\min} = 0.75 \text{ m/s}$ (to prevent vegetation growth)



Design of Earthen Channels

A. Maximum permissible velocity method

B. Tractive force method

General Guideline:

Froude Number should be less than **0.35**



Regular and Most Efficient Cross Sections

A. Regular Cross-Section

$$Q = VA$$

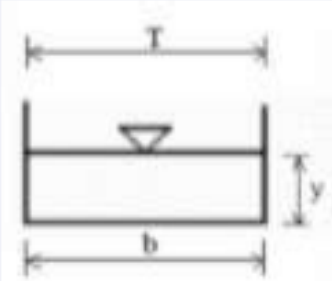
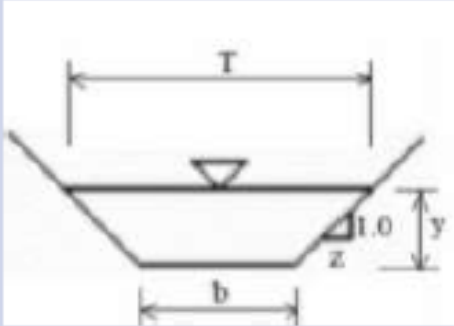
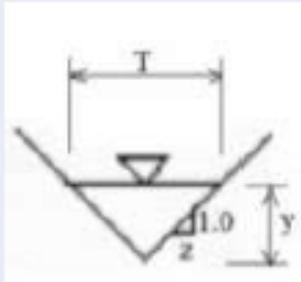
$$Q = \frac{1}{n} R^{2/3} S^{1/2} A$$

$$\frac{Qn}{S^{1/2}} = (A/P)^{2/3} A$$

$$\frac{Qn}{S^{1/2}} = \frac{A^{5/3}}{P^{2/3}}$$



CHANNEL GEOMETRY

	Flow Area (A)	Wetted perimeter (P)	Hydraulic Radius ($R=\frac{A}{P}$)	Top Width (T)	Hydraulic Depth ($D=\frac{A}{T}$)
	by	$b + 2y$	$\frac{by}{b + 2y}$	b	y
	$(b + zy)y$	$b + 2y\sqrt{1 + z^2}$	$\frac{(b + zy)y}{b + 2y\sqrt{1 + z^2}}$	$b + 2zy$	$\frac{(b + zy)y}{b + 2zy}$
	zy^2	$2y\sqrt{1 + z^2}$	$\frac{zy^2}{2y\sqrt{1 + z^2}}$	$2zy$	$\frac{1}{2}y$
	$\frac{2}{3}Ty$	$T + \frac{8y^2}{3T}$	$\frac{2T^2y}{2T^2 + 8y^2}$	$\frac{3A}{2y}$	$\frac{2}{3}y$

Sample Question:

Ratio of the horizontal to vertical dimension of the channel wall

- a) Slope
- b) Channel gradient
- c) Side slope
- d) All of the above

Sample Question 31:

Water flows uniformly in a **rectangular**, concrete, open channel that is 10 m wide at a depth of 3 m. The channel slope is 0.0025. Using $n=0.013$, find the **velocity** (m/s), **flow rate** (m^3/s), and **boundary shear stress** (Pa).

5.848 m/s, 175.44 m^3/s , 45.98 Pa

Solution:

$$v = \frac{1}{n} R^{2/3} S^{1/2}$$

$$A = 10(3) = 30 \text{ m}^3$$

$$P = 10 + 3(2) = 16 \text{ m}^3$$

$$R = A/P = 30/16 = 1.875 \text{ m}$$

$$v = \frac{1}{0.013} 1.875^{2/3} 0.0025^{1/2}$$

$$v = 5.848 \text{ m/s}$$

$$[Q = Av]$$

$$Q = 30(5.848)$$

$$Q = 175.45 \text{ m}^3/\text{s}$$

$$[t = \frac{1}{v} RS]$$

$$t = 9810(1.875)(0.0025)$$

$$t = 45.98 \text{ Pa}$$

Sample Question 32:

A **rectangular**, concrete channel, 15 m wide is to carry water at the rate $22 \text{ m}^3/\text{s}$. If the channel slope is 0.00025, determine the **normal depth** of flow. Use roughness coefficient $n=0.013$.

$$Q = A \frac{1}{n} R^{2/3} S^{1/2}$$

$$A = 15d$$

$$R = \frac{A}{P} = \frac{15d}{15+2d}$$

$$22 = 15d \frac{1}{0.013} \left[\frac{15d}{15+2d} \right]^{2/3} (0.00025)^{1/2}$$

$$d = 1.187 \text{ m}$$

$$1.187 \text{ m}$$

Sample Question 33:

Determine the uniform **flow (m^3/s)** through a **trapezoidal** concrete lined canal having side slope of 3H to 4V and bottom width of 2 m if the depth of flow is 2m. The channel is laid on a slope of 3 m per 2 kilometers. Use $n=0.013$.

$20.85 m^3/s$

Solution:

$$Q = A \frac{1}{n} R^{2/3} S^{1/2}$$

$$A = by + zy^2 = (2)(2) + \left(\frac{3}{4}\right)(2)^2 = 7m^2$$

$$P = b + 2y\sqrt{1 + z^2} = 2 + (2)(2)\sqrt{1 + \left(\frac{3}{4}\right)^2} = 7m$$

$$R = \frac{A}{P} = \frac{7}{7} = 1$$

$$S = \frac{3}{2000} = 0.0015$$

$$Q = (7) \frac{1}{0.013} (1)^{2/3} (0.0015)^{1/2}$$

$$Q = 20.85 \text{ m}^3/\text{s}$$

Sample Question 34:

A **trapezoidal** channel has a bottom width of 6 m and side slopes of 2 horizontal to 1 vertical. If the depth of flow is 1.2 m and the flow is $20.40 \text{ m}^3/\text{s}$, compute

- (a) The specific energy
- (b) Slope of the channel if $n=0.014$
- (c) Average shearing stress at the boundary

1.409 m, 0.00094, 8.18 Pa

Solution:

$$Q = 20.4 \text{ m}^3/\text{s}$$

$$A = 6(1.2) + 2(1.2)^2 = 10.08 \text{ m}^2$$

$$[v = \frac{Q}{A}]$$

$$v = \frac{20.4}{10.08} = 2.0238 \text{ m/s}$$

$$\begin{aligned} \text{(a) Specific energy, } H &= \frac{v^2}{2g} + d \\ &= \frac{2.0238^2}{2(9.81)} + 1.2 \end{aligned}$$

$$\text{Specific energy, } H = 1.409 \text{ m}$$

(b) Slope

$$v = \frac{1}{n} R^{2/3} S^{1/2}$$

$$R = \frac{A}{P} = \frac{10.08}{6+2(2.683)}$$

$$R = 0.8868 \text{ m}$$

$$2.0238 = \frac{1}{0.014} (0.8868)^{2/3} S^{1/2}$$

$$S = 0.00094$$

(c) Shearing Stress:

$$S_s = \gamma R S$$

$$= 9.81(0.8868)(0.00094)$$

$$= 0.00818 \text{ kPa}$$

$$S_s = 8.18 \text{ Pa}$$

Sample Question 35:

Water flows in a **triangular** V-notch steel channel, with vertex angle of 60° , at a velocity of 1.2 m/s. Find the **normal depth** of flow if the channel is laid on a slope of 0.0017. Use $n=0.014$.

$$v = \frac{1}{n} R^{2/3} S^{1/2}$$

$$A = d^2 \tan\left(\frac{\theta}{2}\right)$$

$$P = \frac{2d}{\cos\left(\frac{\theta}{2}\right)}$$

$$R = \frac{A}{P} = \frac{d^2 \tan\left(\frac{60}{2}\right)}{\frac{2d}{\cos\left(\frac{60}{2}\right)}} = 0.25d$$

$$1.2 = \frac{1}{0.014} (0.25d)^{2/3} (0.0017)^{1/2}$$

$$d = 1.04 \text{ m (normal depth)}$$

1.04 m

Sample Question:

What is the depth of water in a trapezoidal channel with a side slope of 2 and carrying a $3.2 \text{ m}^3/\text{s}$ water flow? The channel's bottom width is 1.5 meters and the flowing water has a velocity of 0.85 m/s .

- a) 1.8 m
- b) 1.79 m
- c) 1.05 m
- d) 1.04 m

$$Q = AV \text{ or } A = \frac{Q}{V}$$

$$A = \frac{3.2 \text{ m}^3}{0.85 \text{ m/s}} = 3.765 \text{ m}^2$$

$$A = by + zy^2$$

$$3.765 \text{ m}^2 = 1.5 y + 2y^2$$

$$y = 1.05 \text{ m}$$

C

Sample Question:

What is the depth of water in a trapezoidal channel with a side slope of 2 and carrying a $2.5 \text{ m}^3/\text{s}$ water flow? The channel's bottom width is 1.5 meters and the flowing water has a velocity of 0.8 m/s .

- a) a. 1 m
- b) b. 1.2 m
- c) c. 0.93 m
- d) d. 0.82 m

$$Q = AV \text{ or } A = \frac{Q}{V}$$

$$A = \frac{2.5 \text{ m}^3}{0.8 \text{ m/s}} = 3.125 \text{ m}^2$$

$$A = by + zy^2$$

$$3.125 \text{ m}^2 = 1.5 y + 2y^2$$

$$y = 0.93 \text{ m}$$

C

Sample Question:

A trapezoidal channel has a base width of 6 meters and a side slope of 1H:1V. The channel bottom slope is 0.0002 and the Manning roughness coefficient is 0.014. What is the depth of flow if $Q = 12.1 \text{ m}^3/\text{s}$?

- a) 1.5 m
- b) 1.3 m
- c) 1.7 m
- d) 1.1 m

$$A = by + zy^2$$

$$P = b + 2y\sqrt{1 + z^2}$$

$$Q = A \frac{1}{n} R^{2/3} S^{1/2}$$

$$12.1 = (6y + 1y^2) \frac{1}{0.014} \left[\frac{(6y + 1y^2)}{6 + 2y\sqrt{1 + 1^2}} \right]^{2/3} 0.0002^{1/2}$$

$$y = 1.5 \text{ m}$$

B. Best Hydraulic Section

For a constant AREA,

- Discharge is MAXIMUM
- Velocity is MAXIMUM

$$Q=VA$$

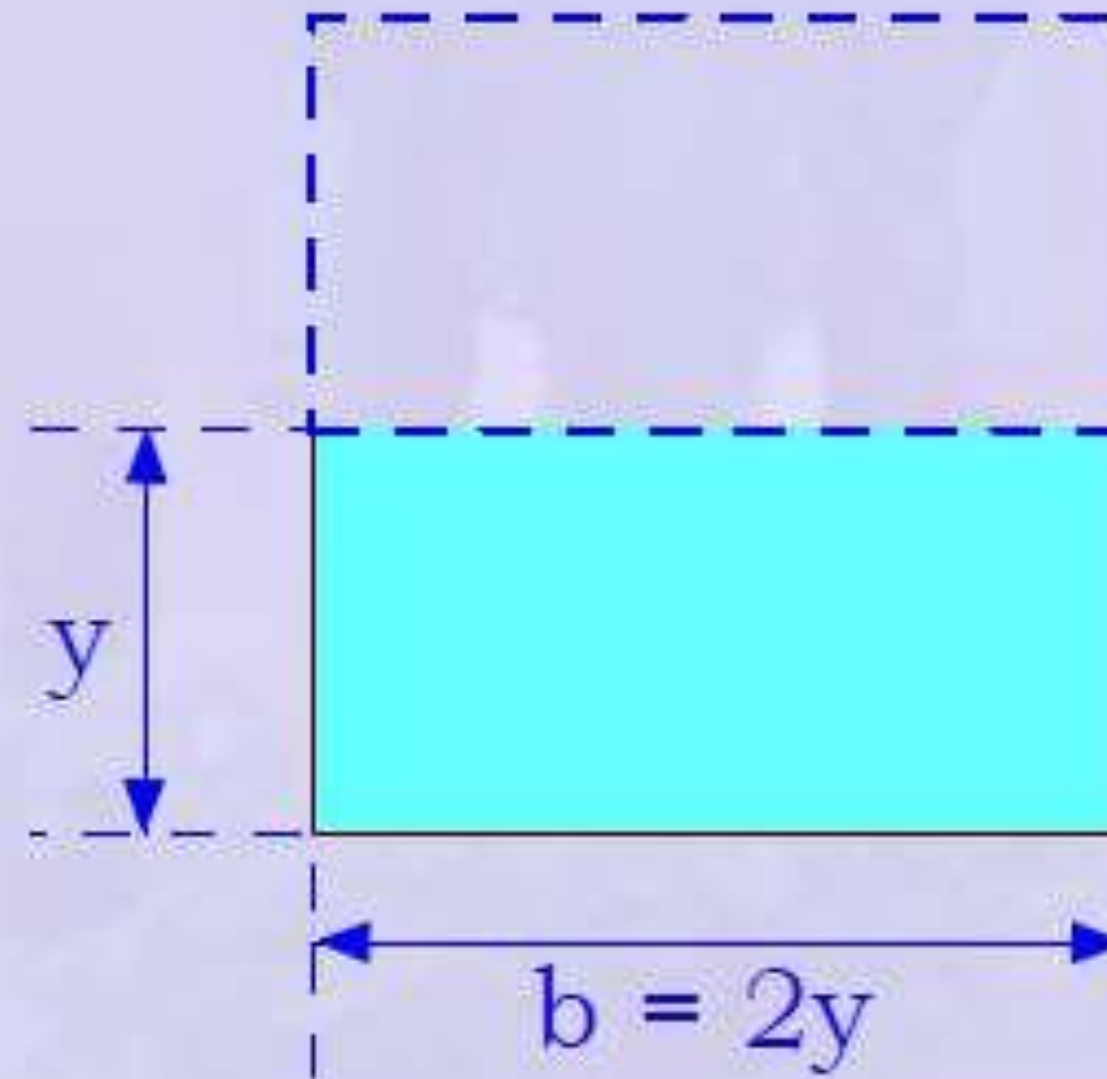
$$V = \frac{1}{n} R^{2/3} S^{1/2}$$

- Hydraulic Radius is MAXIMUM
- Perimeter is MINIMUM

$$R=A/P$$



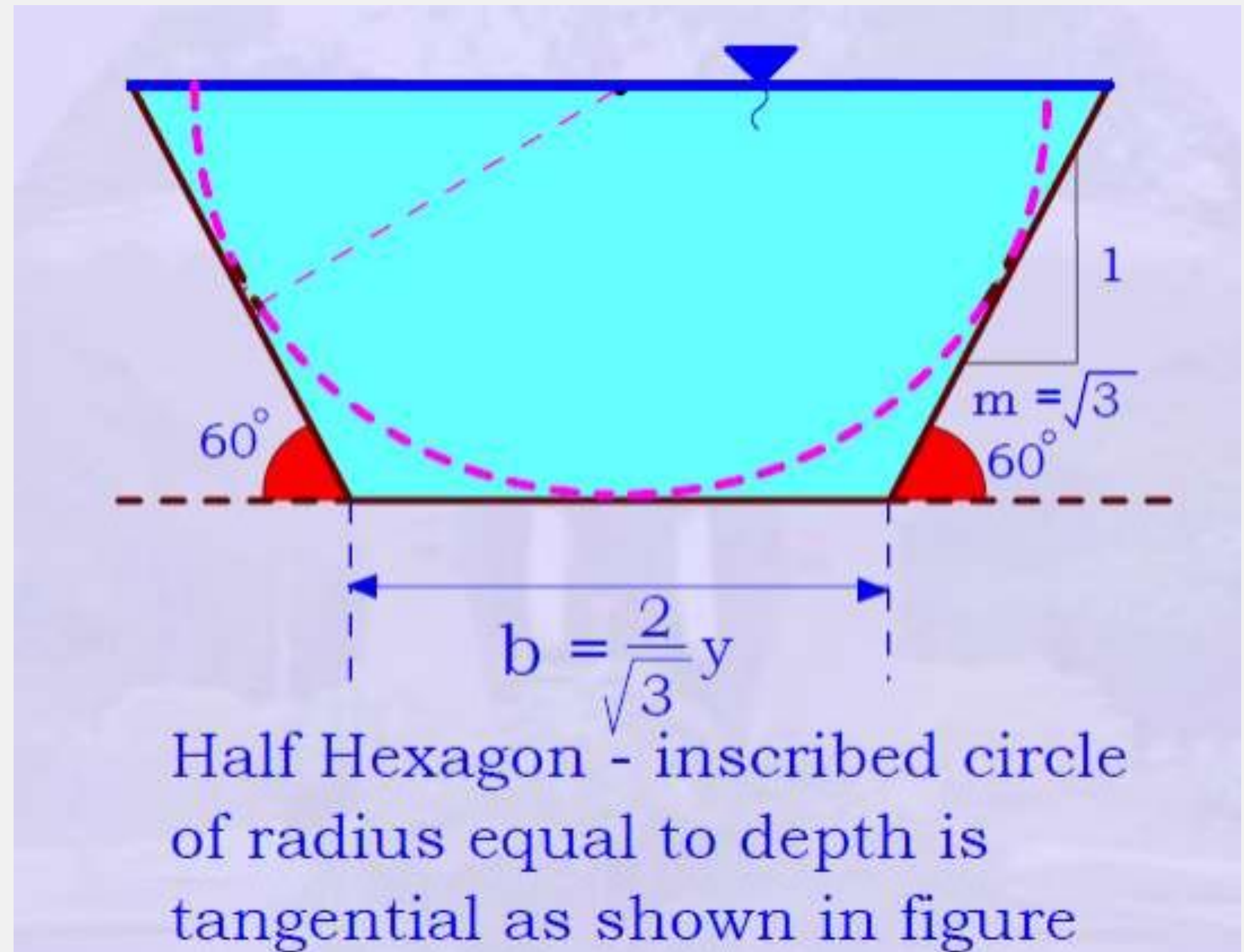
Best Hydraulic Section for **RECTANGLE**



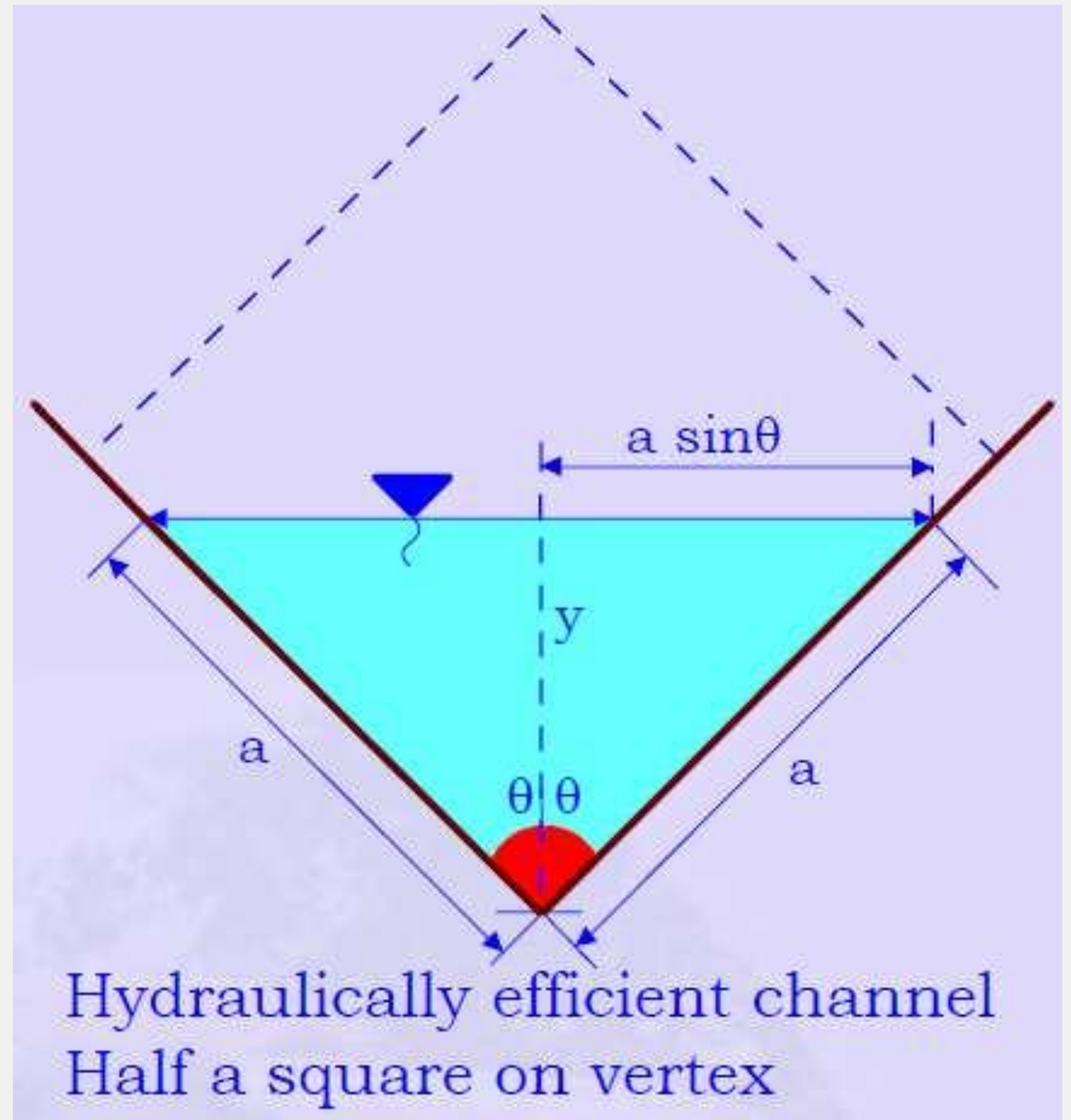
Hydraulically efficient rectangular channel is half of a square.



Best Hydraulic Section for TRAPEZOID



Best Hydraulic Section for **TRIANGLE**



Best Hydraulic Channel Section Properties

Cross sectional Shape	Area, A	Wetted Perimeter, P	Hydraulic Radius, R	Top Width, T
Rectangular (half of a square) $\theta = 90^\circ$	$2d^2$	$4d$	$0.5d$	$2d$
Trapezoidal (half of a hexagon) $\theta = 60^\circ$	$\sqrt{3}d^2$	$2\sqrt{3}d$	$0.5d$	$\frac{4\sqrt{3}}{3}d$
Triangular (half of a square on vertex) $\theta = 45^\circ$	d^2	$2\sqrt{2}d$	$\frac{\sqrt{2}}{4}d$	$2d$
Semi Circular	$\frac{\pi}{2}d^2$	πd	$0.5d$	$2d$



BEST HYDRAULIC CROSS-SECTION

(best b-d proportion)

Concrete (Lined) Canals	Unlined Canals
$b = 2 d \tan(\theta/2)$	$b = 4 d \tan(\theta/2)$
Top Width (t) at water surface level d: $t = b + (2d/\tan\theta)$ Total Top Width (T) at total depth D level: $T = b + (2D/\tan\theta)$	



Sample Question 36:

What dimensions would you recommend for a canal that will carry $8.5 \text{ m}^3/\text{s}$ of water with a velocity of 2.0 m/s considering the best hydraulic section for (a) rectangle, (b) trapezoid, (c) triangle, and (d) semicircle?

1.46 m, 1.57 m, 2.06 m, 1.65 m

Solution to Question 36:

$$Q = V/A = 8.5/2 = 4.25 \text{ m}^2$$

For rectangle:

$$A = 2d^2$$

$$4.25 = 2d^2$$

$$d = 1.46 \text{ m}$$

For trapezoid:

$$A = \sqrt{3}d^2$$

$$4.25 = \sqrt{3}d^2$$

$$d = 1.57 \text{ m}$$

For triangle:

$$A = d^2$$

$$4.25 = d^2$$

$$d = 2.06 \text{ m}$$

For semicircle:

$$A = \frac{\pi}{2} d^2$$

$$4.25 = \frac{\pi}{2} d^2$$

$$d = 1.645 \text{ m}$$

1.46 m, 1.57 m, 2.06 m, 1.65 m

Sample Question 37:

An open channel is to be designed to carry $1.5 \text{ m}^3/\text{s}$ at a slope of 0.007. The channel material has an n value of 0.012. Find the **most efficient depth and top width** for **semicircular** section.

$$d = 0.564 \text{ m}, T = 1.128 \text{ m}$$

Solution:

$$Q = A \frac{1}{n} R^{2/3} S^{1/2}$$

For semicircular section:

$$A = \frac{\pi}{2} d^2 \quad ; \quad R = 0.5d$$

$$AR^{2/3} = \frac{Qn}{\sqrt{S}}$$

$$AR^{2/3} = \frac{1.5(0.012)}{\sqrt{0.007}}$$

$$AR^{2/3} = 0.215$$

$$\left(\frac{\pi}{2} d^2\right) (0.5d)^{2/3} = 0.215$$

$$0.99 d^{8/3} = 0.215$$

$$d = \mathbf{0.564 \text{ m}}$$

$$T = 2d = 2(0.564 \text{ m}) = \mathbf{1.128 \text{ m}}$$

Sample Question 38:

An open channel is to be designed to carry $1.5 \text{ m}^3/\text{s}$ at a slope of 0.007. The channel material has an n value of 0.012. Find the **most efficient depth and top width** for rectangular section.

$$d = 0.515 \text{ m}, T = 1.03 \text{ m}$$

Solution:

For most efficient rectangular section:

$$b = 2d$$

$$A = 2d^2$$

$$R = 0.5d$$

Then;

$$1.5 = (2d^2) \frac{1}{0.012} (0.5d)^{\frac{2}{3}} (0.007)^{\frac{1}{2}}$$

$$d^{8/3} = 0.1707$$

$$d = 0.515 \text{ m}$$

$$b = 2d = 1.03 \text{ m}$$

Sample Question 39:

A trapezoidal channel carrying 400 cfs is built with non-erodible bed having a slope of 0.0016 and $n = 0.025$. Determine the depth of water y if $b = 20$ ft and $z = 2$.

- a) 3.36 ft
- b) 2.87 ft
- c) 5.72 ft
- d) 4.13 ft

A

Solution to Question 39:

For English units:

$$Q = A \frac{1.486}{n} R^{2/3} S^{1/2}$$

$$AR^{2/3} = \frac{Qn}{1.486\sqrt{S}}$$

$$AR^{2/3} = \frac{400(0.025)}{1.486\sqrt{0.0016}} = 168.2$$

Where:

$$A = by + zy^2 ; \quad R = \frac{by + zy^2}{b + 2y\sqrt{1 + z^2}}$$

$$(by + zy^2) \left(\frac{by + zy^2}{b + 2y\sqrt{1 + z^2}} \right)^{2/3} = 168.2$$

$$\frac{(by + zy^2)^{5/3}}{(b + 2y\sqrt{1 + z^2})^{2/3}} = 168.2$$

$$\frac{(20y + 2y^2)^{5/3}}{(20 + 2y\sqrt{1 + 2^2})^{2/3}} = 168.2$$

$$y = 3.36 \text{ ft}$$

A

Sample Question 40:

On previous problem, determine the cross-sectional area, A of the channel.

- a) 72.36 ft^2
- b) 89.78 ft^2
- c) 63.56 ft^2
- d) 83.92 ft^2

$$A = 20(3.36) + 2(3.36)^2 = 89.78 \text{ ft}^2$$

B

Sample Question 41:

On previous problem, determine the best hydraulic section?

- a) $y = 3.36$ ft; $b = 3.74$ ft
- b) $y = 4.52$ ft; $b = 6.26$ ft
- c) $y = 6.60$ ft; $b = 7.62$ ft
- d) $y = 7.81$ ft; $b = 9.76$ ft

For most efficient trapezoidal section,
 $A = \sqrt{3}d^2; R = 0.5d$

$$\begin{aligned}\text{Since } AR^{2/3} &= 167.7 \\ (\sqrt{3}d^2)(0.5d)^{2/3} &= 168.2 \\ 1.09d^{8/3} &= 168.2 \\ \mathbf{d} &= \mathbf{6.6\ ft}\end{aligned}$$

$$\begin{aligned}A &= \sqrt{3}d^2 = \sqrt{3}(6.6)^2 = 75.45\ ft^2 \\ A &= bd + zd^2 \\ 75.45 &= b(6.6) + 0.577(6.6)^2 \\ \mathbf{b} &= \mathbf{7.62\ ft}\end{aligned}$$

C

Sample Question:

A triangular channel with most efficient proportion discharges water at the rate of 1 m³/s. Assuming $n=0.018$ and $S=0.0021$ calculate the normal depth of flow in meters.

$$A = d^2$$

$$R = \frac{\sqrt{2}}{4} d = 0.3536 d$$

$$AR^{2/3} = \frac{Qn}{\sqrt{S}}$$

$$d^2 (0.3536 d)^{\frac{2}{3}} = \frac{1(0.018)}{\sqrt{0.0021}}$$

$$0.5d^{8/3} = 0.39279$$

$$d = 0.913 \text{ m}$$

0.913 m

Sample Question:

Determine the **top width** at water surface level of the most efficient trapezoidal concrete open channel if the design depth is 5 meters, design discharge is 100 m³/s and velocity is 2 m/s. Also determine the **total top width if 15% freeboard is applied**.

Use the Design Criteria: Most Efficient Canal ($\Theta=60^\circ$ and $z = 0.577$)

$$Q = AV$$

$$100 = A (2)$$

$$A = 50 \text{ m}^2$$

$$A = bd + zd^2$$

$$A - zd^2 = bd$$

$$b = (A - zd^2) / d$$

$$= [50 - 0.577(5)^2] / 5$$

$$= 7.12 \text{ m}$$

$$\begin{aligned} t &= b + (2d / \tan \Theta) \\ &= 7.12 \text{ m} + [2(5) / 1.732] \\ &= 12.89 \text{ m} \end{aligned}$$

$$\begin{aligned} D &= 1.15d \\ &= 1.15(5) \\ &= 5.75 \text{ m} \end{aligned}$$

$$\begin{aligned} T &= b + (2D / \tan \Theta) \\ &= 7.12 + [2(5.75) / 1.732] \\ &= 7.12 + 6.64 \\ &= 13.8 \text{ m} \end{aligned}$$

12.89 m, 13.8 m

Sample Question:

What is the **base** of the most efficient trapezoidal concrete open channel if the discharge is $100 \text{ m}^3/\text{s}$ and the velocity is 2 m/s ?

$$A = \frac{Q}{V} = \frac{100}{2} = 50 \text{ m}^2$$

$$A = \sqrt{3}y^2$$

$$50 = \sqrt{3}y^2$$

$$y = 5.4 \text{ m}$$

$$A = by + zy^2$$

For most efficient trapezoidal channel,

$$z = \frac{1}{\tan(60)} = 0.577$$

$$50 = b(5.4) + 0.577(5.4)^2$$

$$b = 6.14 \text{ m}$$

6.14 m

Sample Question:

Estimate the **width** and the **depth** of concrete-lined rectangular open channel for water velocity of 2 m/s and discharge of 10 m³/s.

$$A = \frac{Q}{V} = \frac{10}{2} = 5 \text{ m}^2$$

$$A = bd$$

$$5 = (2d)d$$

$$d = 1.58 \text{ m} \approx 1.6 \text{ m}$$

$$b = 2d = 2(1.58) = 3.16 \text{ m} \approx 3.2 \text{ m}$$

3.2 m, 1.6 m

Sample Question:

What should be the **depth** and the **side angle** with the horizontal of concrete-lined triangular open channel for a cross-sectional area of 50 m^2 ?

$$A = d^2$$

$$50 = d^2$$

$$d = 7.1 \text{ m}$$

$$\theta = 45^\circ \text{ for efficient triangular canals.}$$

7.07 m, 45°

Sample Question:

What is the **bottom width** for best hydraulic cross-section of unlined open channel with minimum seepage if design depth is 5 meters and side slope is 2:1?

$$\tan\theta = \frac{1}{2}$$

$$\theta = \tan^{-1}\left(\frac{1}{2}\right) = 26.565^\circ$$

$$\begin{aligned} b &= 4 d \tan (\theta/2) \\ &= 4 (5) \tan (26.6/2) \\ &= 4.72 \text{ m} \end{aligned}$$

4.72 m

Sample Question:

If an unlined trapezoidal canal with best hydraulic cross-section can be used, what actual **depth** of open channel would you recommend to carry $10 \text{ m}^3/\text{s}$ with a velocity of 1 m/s ? Use 2:1 side slope and 15% freeboard.

$$A = \frac{Q}{V} = \frac{10}{1} = 10 \text{ m}^2$$

$$\tan\theta = \frac{1}{z}$$

$$\theta = \tan^{-1}\left(\frac{1}{\left(\frac{2}{1}\right)}\right) = 26.565^\circ$$

For unlined channels,

$$b = 4d \tan\left(\frac{\theta}{2}\right) = 4d \tan\left(\frac{26.565}{2}\right) = 0.944 d$$

$$A = bd + zd^2$$

$$10 = (0.944d)d + 2d^2$$

$$d = 1.84 \text{ m}$$

$$D = 1.15 d = 1.15(1.84) = \mathbf{2.12 \text{ m}}$$

2.12 m

RECALL

TYPES OF FLOW

1. **Steady flow** – flow depth does not change with time
2. **Unsteady flow** – flow depth changes with time
3. **Uniform flow** – flow depth does not change with distance
Normal depth d_n – the depth at which uniform flow will occur in an open channel.
4. **Non-uniform flow** – flow depth varies with distance
 - a. **Gradually varied** – depth of flow changes gradually either as a drawdown curve or backwater curve
 - b. **Rapidly varied** – flow depth changes abruptly e.g. hydraulic jump or hydraulic drop



States of Flow (Froude Number)

$$Fr = \frac{V}{\sqrt{gD}}$$

1. CRITICAL FLOW: $Fr = 1$
2. SUBCRITICAL FLOW: $Fr < 1$
3. SUPERCRITICAL FLOW: $Fr > 1$



CRITICAL DEPTH d_c

- A depth that for a given total specific energy/ total head, the **discharge is maximum**.
- The depth at which for a given flow, the **specific energy is minimum**.

$$Q = A\sqrt{2g(H - d)}$$



Critical Depth on **RECTANGULAR SECTION**

$$d_c = \sqrt[3]{\frac{q^2}{g}}$$

$$q = vd = \frac{Q}{b}$$

q = unit flow in m^3/s per meter width of canal
Q = total flow in m^3/s
B = channel width



Sample Question:

The critical depth in a rectangular channel with a bottom width of 4.0 m for a discharge of 6 m³ /s is

- a. 0.34 m
- b. 0.44 m
- c. 0.54 m
- d. 0.61 m

$$d_c = \sqrt[3]{\frac{q^2}{g}}$$

$$q = \frac{Q}{b}$$

$$d_c = \sqrt[3]{\frac{\left(\frac{6}{4}\right)^2}{9.81}} = 0.61 \text{ m}$$

D

Sample Question:

The critical depth in a rectangular channel with a bottom width of 4.0 m for a discharge of 7 m³ /s is

- a. 0.34 m
- b. 0.44 m
- c. 0.68 m
- d. 0.61 m

$$d_c = \sqrt[3]{\frac{q^2}{g}}$$

$$q = \frac{Q}{b}$$

$$d_c = \sqrt[3]{\frac{\left(\frac{7}{4}\right)^2}{9.81}} = 0.68 \text{ m}$$

c

HYDRAULIC JUMP

- It is an abrupt increase in depth of rapidly flowing water
- Flow at the jump changes from supercritical to subcritical stage with an accompanying loss of kinetic energy.
- It is a classic example of rapidly varied flow



Sample Question 42:

A hydraulic jump is a classic example of this type of flow

- a. Unsteady flow
- b. Gradually varied flow
- c. Rapidly Varied flow
- d. Critical flow

C

Sample Question:

If q is the discharge per unit width of a channel and D_1 and D_2 are the depths of water before and after hydraulic jump, which of the following relationship is true?

A. $D_1 D_2 (D_2 - D_1) = \frac{2q^2}{g}$

C. $D_1 D_2 (D_2 + D_1) = \frac{2q^2}{g}$

B. $\frac{D_2}{D_1} (D_2 - D_1) = \frac{2q^2}{g}$

D. $\frac{D_2}{D_1} (D_2 + D_1) = \frac{2q^2}{g}$

C

Sample Question 43:

Water is flowing at a depth of 1.5 m in a 3-m-wide rectangular channel having $n=0.013$ and $S=0.0009$. (a) Is the flow critical, subcritical, or supercritical (b) Determine the critical depth, and (c) determine the critical slope.

$$F_N = 0.497 < 1.0 \text{ (subcritical)}$$

$$0.94 \text{ m}$$

$$0.00324$$

Solution:

(a) Stage of flow:

$$v = \frac{1}{n} R^{2/3} S^{1/2}$$

$$v = \frac{1}{0.013} \left[\frac{(3)(1.5)}{3+2(1.5)} \right]^{2/3} (0.0009)^{1/2}$$

$$v = 1.905 \frac{m}{s}$$

Froude No,

$$F_N = \frac{v}{\sqrt{gd}} = \frac{1.905}{\sqrt{9.81(1.5)}}$$

$$= \mathbf{0.497 < 1.0 \text{ (subcritical)}}$$

(b) Critical depth

$$q = vd = 1.905(1.5) = 2.8575 \text{ m}^3/\text{s per m}$$

$$dc = 3 \sqrt{\frac{q^2}{g}} = 3 \sqrt{\frac{(2.8575)^2}{9.81}} = \mathbf{0.94 \text{ m}}$$

(c) Critical slope

$$v_c = \frac{q}{d_c} = \frac{2.8575}{0.94}$$

$$v_c = 3.04 \frac{m}{s}$$

$$v = \frac{1}{n} R^{2/3} S^{1/2}$$

$$R_c = \frac{A_c}{P_c} = \frac{3(0.94)}{[3 + 2(0.94)]} = 0.578 \text{ m}$$

$$3.04 = \frac{1}{0.013} (0.578)^{2/3} S_c^{1/2}$$
$$\mathbf{S_c = 0.00324}$$

Sample Question 44:

A trapezoidal channel has a base width of 6 meters and a side slope of 1H:1V. The channel bottom slope is 0.0002 and the Manning roughness coefficient is 0.014. What will be the state of flow if the depth of flow is 1.5 m?

- a) Critical
- b) Sub-critical
- c) Supercritical
- d) Laminar

$$v = \frac{1}{n} R^{2/3} S^{1/2}$$

$$v = \frac{1}{0.014} \left(\frac{6(1.5) + 1(1.5^2)}{6 + 2(1.5)\sqrt{1 + 1^2}} \right)^{2/3} (0.0002)^{1/2} = 1.08 \text{ m/s}$$

$$F_N = \frac{v}{\sqrt{gd}} = \frac{1.08}{\sqrt{9.81(1.5)}} = 0.28(\text{subcritical})$$

B

Sample Question 45:

What is the critical depth for a grassy **triangular** channel with 2H:1V side slopes and a 0.5% slope when the flow is 3.00 m³/s?

$$\frac{A^3}{T} = \frac{Q^2}{g}$$

$$A = zy^2 = 2y^2$$

$$T = 2zy = 2(2)y = 4y$$

$$Q = 3.00 \text{ m}^3/\text{s}$$

$$\frac{(2y^2)^3}{4y} = \frac{3^2}{9.81}$$

$$y = 0.86 \text{ m}$$

0.86 m

